Today:

☆ auxiliary results:

- uniqueness of π for an irreducible Markov chain:

irreducible only has room for one π.

- aperiodic case: follows from convergence theorem.

if $p^n(x,y) \longrightarrow \pi(y) \quad \forall x$ as $n \to \infty$

if $X_0 \overset{d}{=} \nu$ & assume ν is stationary.

Then dist$^{\underline{bn}}$ at time n has

$$P(X_n = y \mid X_0 \overset{d}{=} \nu)$$

$$= \sum_x \nu(x)\, p^n(x,y)$$

which $\xrightarrow{n\to\infty} \sum_x \nu(x)\pi(y)$

$= \pi(y)\cdot 1$

but if stationary then

$$P(X_n = y \mid X_0 \overset{d}{=} \nu) = \nu(y)$$

$\therefore \quad \nu(y) = \pi(y) \quad \forall y.$

- periodic case: the lazy Markov chain.

ε-lazy chain of trans. mtx. p

has trans. mtx

$$q := \varepsilon I + (1-\varepsilon)p$$

wp ε do nothing

wp $1-\varepsilon$ follow matrix p.

$\forall \varepsilon > 0$, p & q have same stat. dist.

if $\pi = \pi p$

then $\pi = \pi q$

& vice-versa (check)

& if $\varepsilon > 0$, $\forall x$, $q(x,x) > 0$

$\Rightarrow$ Period$(x) = 1$.

(w.r.t. q)

This gives uniqueness

for irreducible MC

whether periodic or aperiodic.

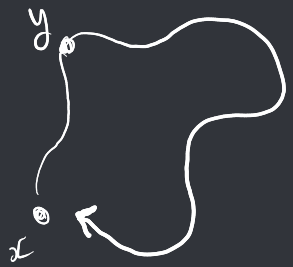
- existence: if state x in a MC has

$$E_x[T_x^+] < \infty \text{ then}$$

$$\exists \text{ stat. dist. } \pi : \pi(x) = \frac{1}{E_x[T_x^+]} > 0.$$

Pf: construct π by letting

$$\pi(y) = \frac{E_x[N_{T_x^+ - 1}(y)]}{E_x[T_x]}$$



visits to y
before returning to x.

$\pi(y)$ = average fraction of one excursion that is spent at y.

proof in book.

sketch: need to show $\pi = \pi p$

entrywise $\pi(v) = \sum_u \pi(u) p(u,v)$ (*)

handle separately $v = x$ & $v \neq x$.

$v \neq x$: count edge crossings & add them up. $\quad u \to v \quad \forall u$

Upon n^{th} visit to u, prob(cross $u \to v$) $= p(u,v)$

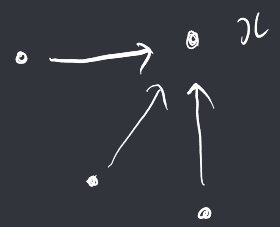
$\therefore \quad E[\#(u,v) \text{ crossings}] = E[\#\text{visits to } u]\, p(u,v)$

$$E[\#\text{visits to } v] = \sum_u E[\#(u,v) \text{ crossings}]$$

$$\therefore \quad E[\#\, v \text{ visits}] = \sum_u E[\#\, u \text{ visits}]\, p(u,v)$$

divide by $E_x[T_x]$ to get $(*)$

$v = x$:



$$\sum_u \#\left(\begin{array}{c}(u,x) \text{ crossings} \\ \text{till return}\end{array}\right) = 1$$

& check its expected value $= \sum_u E[\#(u \text{ visits})]\, p(u,x)$.

Example of MC w/ x recurrent but

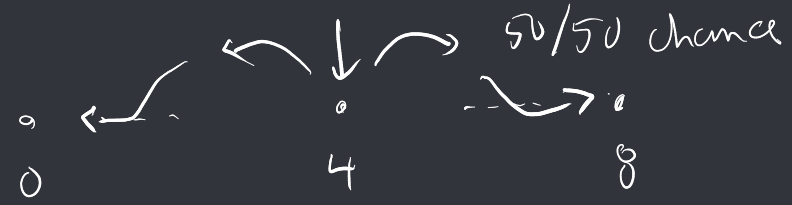
$$E_x[T_x] = \infty.$$

simple random walk on $\mathbb{Z}$.

$$p(x, x+1) = p(x, x-1) = 1/2.$$

$\cdots \quad -2 \quad -1 \quad 0 \quad 1 \quad 2 \quad \cdots$

show 0 recurrent: each time MC reaches state $\pm 2^n$, equally likely to hit 0 or $\pm 2^{n+1}$ first.



Let $N = \max\{n : |X_m| = 2^n \text{ some } m < T_0^+\}$.

Then strong Markov property implies

$$P(N \geq n+1 \mid N \geq n) = \frac{1}{2}$$

$$\& \quad P(N \geq 0) = 1$$

so $$P(N \geq n) = 2^{-n} \xrightarrow{n \to \infty} 0$$

$\therefore$ will hit 0 before $\to \pm\infty$.

0 is recurrent. No stat. dist:

if there was, $\pi = \pi p$ implies

$$\underbrace{\pi(x) = \pi(x-1)/2 + \pi(x+1)/2}_{\text{line w/ constant slope.}}$$

if slope $\neq 0$, $\pi(x) < 0$ some x. $\Rightarrow\Leftarrow$

$\therefore$ slope $= 0$ so $\pi(x) = c \ \forall x$.

can't make $\sum_x \pi(x) = 1$.

can't normalize $\Rightarrow$ no stat. dist.

☆ <u>example</u>: coupling time.

Random walk on m-digit binary string.

$$(a_1, \ldots, a_m) \in \{0,1\}^m$$

trans: pick $i \in \{1, \dots, m\}$ uniformly.
randomize i^{th} digit (pick 0 or 1 for new value each w.p. $1/2$).

ex: $m=3$

$$(0,1,0) \longrightarrow (0,1,1) \longrightarrow (0,1,1) \longrightarrow \cdots$$

$$\begin{array}{ccc} \uparrow & & \uparrow \\ 1 & & 0 \end{array}$$

Start two copies in arbitrary configurations X_0, Y_0. Flip same digits to same values in both. How long until guaranteed $X_n = Y_n$ regardless of X_0, Y_0?

Know that $X_n = Y_n \quad \forall\, n \geq T$

where

$$T := \min\{n : \text{all entries have been selected} \geq \text{once}\}.$$

How to estimate T?

Let

$$K_n = \{\#\text{ distinct entries flipped up to time } n\}$$

$$T_k = \inf\{n : K_n = k\} \qquad (T_0 = 0)$$

Then $(T_k - T_{k-1})_{k=1}^{n}$ independent

& $T_k - T_{k-1} \overset{d}{=} \text{geo}\left(\frac{m-(k-1)}{m}\right)$

$$\text{success prob} = \frac{\#\text{new entries}}{\text{total } \# \text{ entries}}$$

In particular $\quad E\left[T_{k+1} - T_k\right] = \frac{m}{m-k}$

and since

$$T = \sum_{k=0}^{m-1} \left(T_{k+1} - T_k\right),$$

$$E[T] = \frac{m}{m} + \frac{m}{m-1} + \cdots + \frac{m}{1}$$

$$= m\left(1 + \tfrac{1}{2} + \cdots + \tfrac{1}{m}\right)$$

$$\approx m \log m$$

Specific Markov chains for which stationary distribution easier to compute:

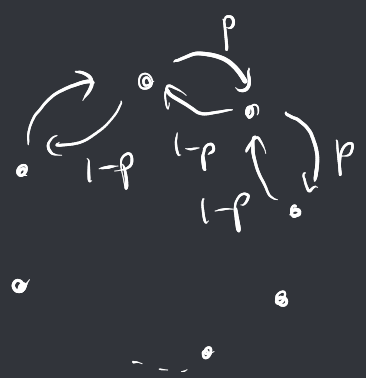
- doubly stochastic ... pertains to Markov chains over a group.
- reversible ... pertains to Hamiltonian systems thermodynamics

MC is doubly stochastic if $\forall y$,

$$\sum_x p(x,y) = 1$$

at any state, sum of incoming probab. $= 1$.

ex:



$$X_{n+1} = \begin{cases} X_n + 1 \bmod m & \text{w.p. } p \\ X_n - 1 \bmod m & \text{w.p. } 1-p \end{cases}$$

then at any i,

$i-1 \to i$ w.p. p

$(\forall k \in \mathbb{Z}$, "$k$ mod m"

$= k - jm$ where j:

$k - jm \in \{0, \dots, m-1\}.)$

ex: $5 \bmod 2 = 1$

odd mod $2 = 1$

even mod $2 = 0$.

$10 \bmod 4 = 2$ ...

$i+1 \longrightarrow i$ w.p. $1-p$

so incoming probabs. sum to 1.

Fact: doubly stochastic finite MC has uniform stat. dist.

i.e. π defined by

$$\pi(x) = \frac{1}{|S|} \quad \forall x \in S$$

is stationary.

Pf. with π as above, $\pi = \pi p$ gives, $\forall y$,

$$\frac{1}{|S|} = \sum_x \frac{1}{|S|} p(x,y)$$

which holds iff $\sum_x p(x,y) = 1$.

<u>ex:</u>

the random n-digit MC is doubly stochastic (check incoming probabs to each state sums to 1).

<u>OR</u> check $p(x,y) = p(y,x)$ $\forall x, y$ which implies doubly stochastic.

ex: if G is a group, $H \subset G$ &

$(p_h)_{h \in H}$ non-negative & $\sum_{h \in H} p_h = 1$

then MC with

$$p(x, hx) = p_h \quad \forall h \in H$$

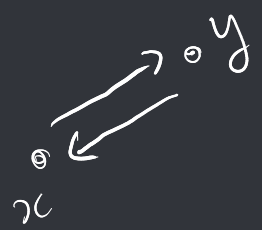
& $p(x,y) = 0$ if $y \neq hx$, some $h \in H$

is doubly stochastic.

Reversible MC

detailed balance holds for a stat. dist. π if $\forall x, y$

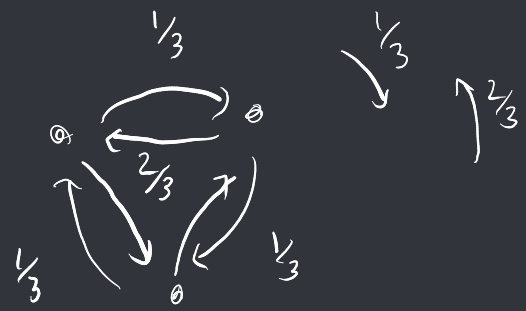
$$(2) \qquad \pi(x)p(x,y) = \pi(y)p(y,x)$$



equivalently, π is reversible when this holds.

(2) says edge crossings have same frequency in both directions.

<u>Q:</u>



is unif. distbtn. reversible?

prefers ↑ over ↓.

should not be. Indeed with adjacent x, y

$$\pi(x)p(x,y) = \frac{1}{|S|} \cdot \frac{1}{3}$$

$$\pi(y)p(y,x) = \frac{1}{|S|} \cdot \frac{2}{3}$$

<u>not</u> reversible.

More generally, if π is stationary for p,

define time-reversed trans. mtx. $\tilde{p}$ by imposing

$$\pi(x)\tilde{p}(x,y) = \pi(y)\,p(y,x)$$

giving

$$\tilde{p}(x,y) = p(y,x)\,\pi(y)/\pi(x)$$

(exercise: verify $\sum_y \tilde{p}(x,y) = 1 \quad \forall x$)

(use $\pi = \pi p$)

if $X_0 \overset{d}{=} \pi$ & n is fixed the

distribution of $(X_n, X_{n-1}, X_{n-2}, \ldots)$

$:= (Y_0, Y_1, Y_2)$

has $Y_0 \overset{d}{=} \pi$ & (Y_n) follows MC

w/ trans. mtx. $\tilde{p}$.